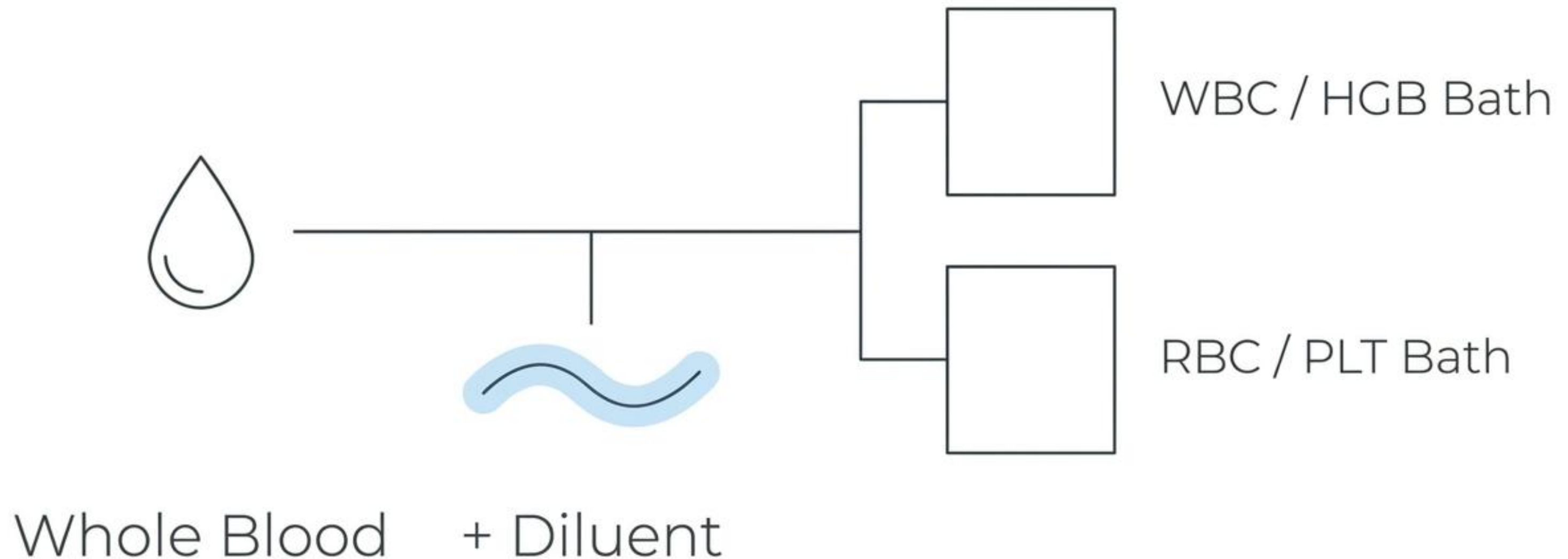


CBC Principles

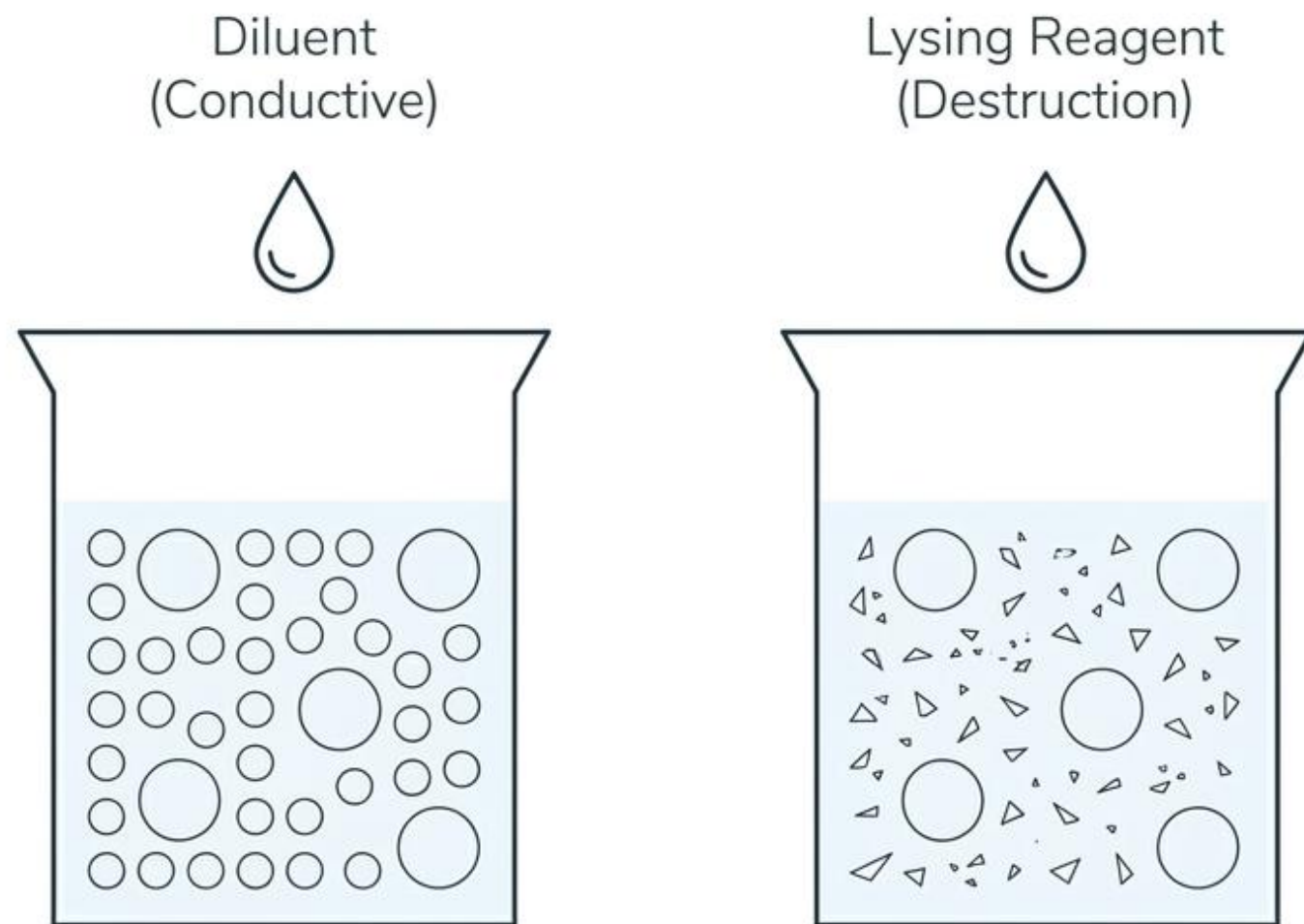
Phases of Laboratory Testing



Automated Fluidic Pathway

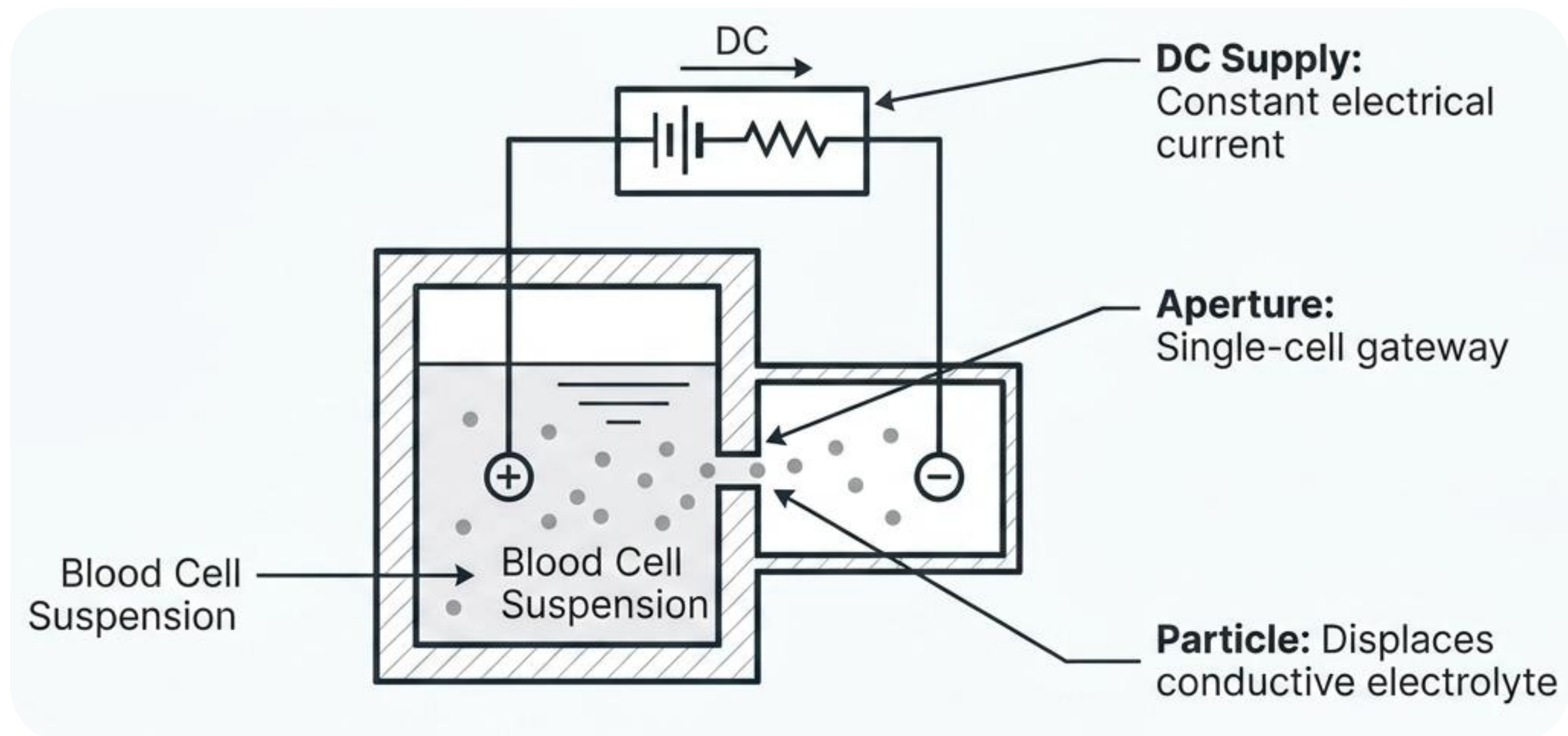


Reagent Systems & Chamber Chemistry

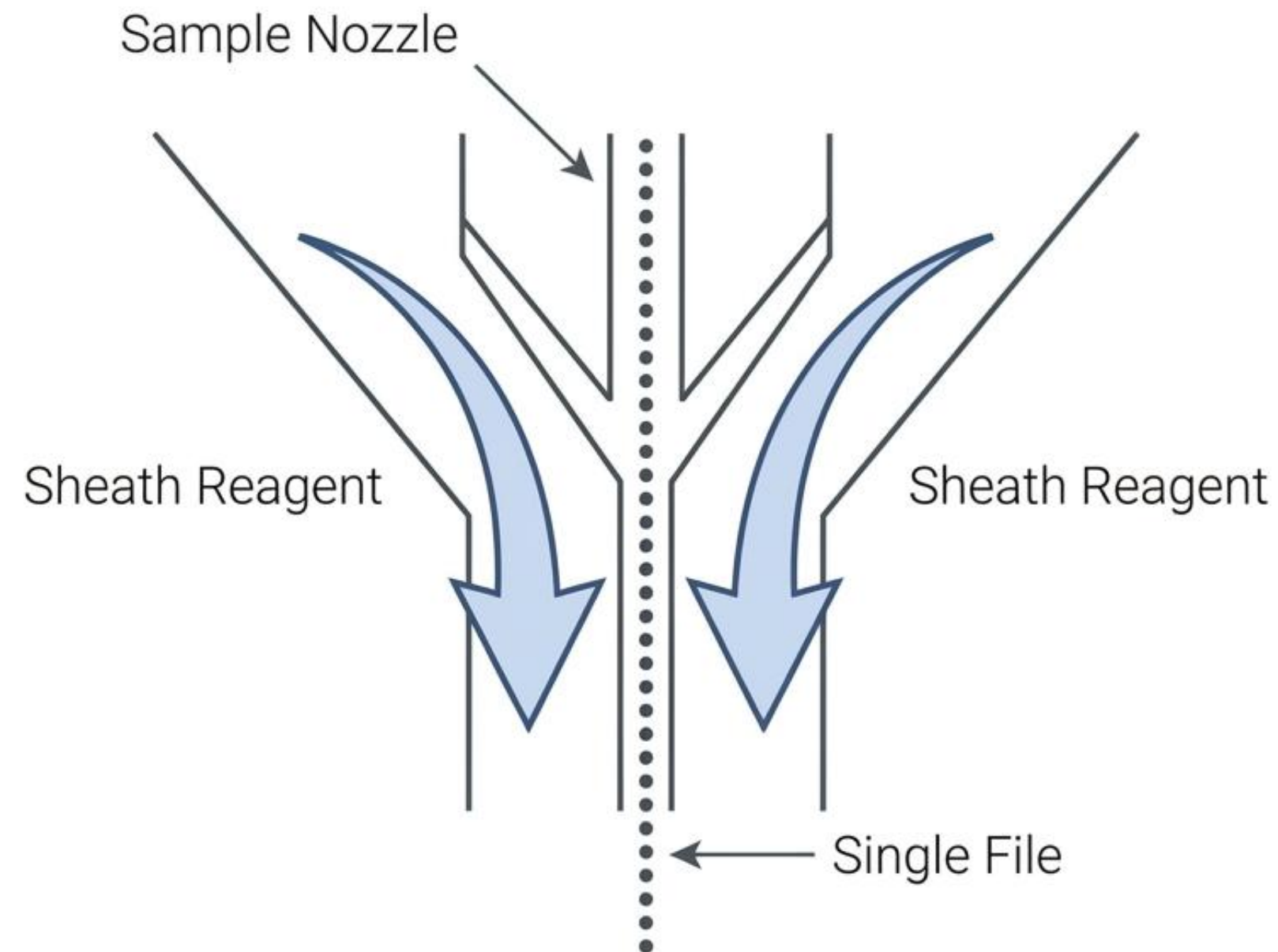


Diluent	Lysing Reagent
Conductive liquid, suspends cells	Destroys cellular membranes
Maintains RBC/PLT integrity	Strips RBCs, releases HGB pigment

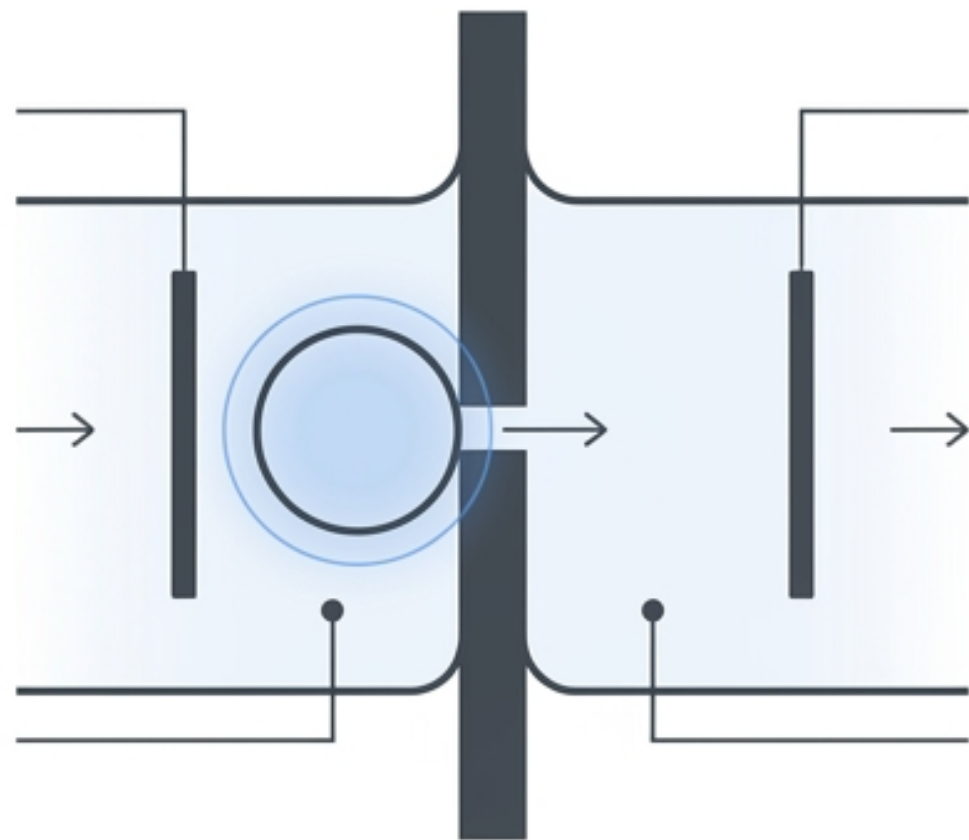
Coulter Principle: Electrical Impedance



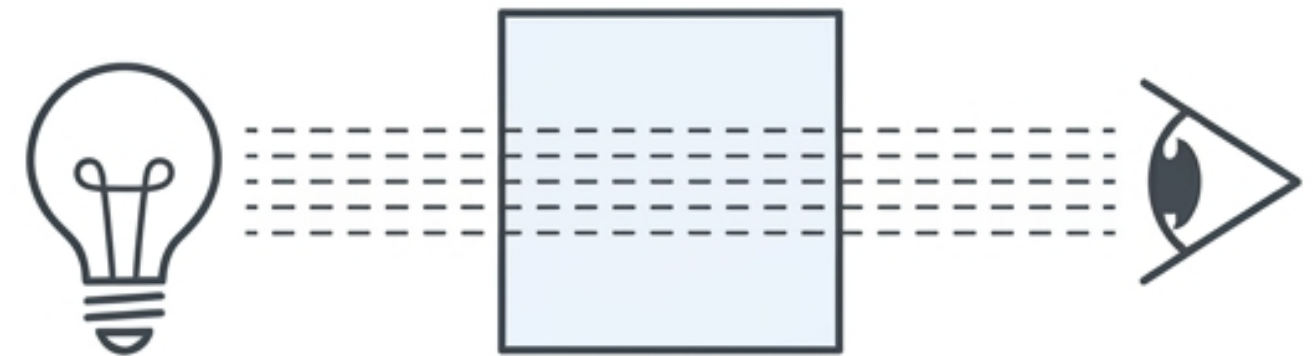
Hydrodynamic Focusing Technology



WBC & HGB Chamber

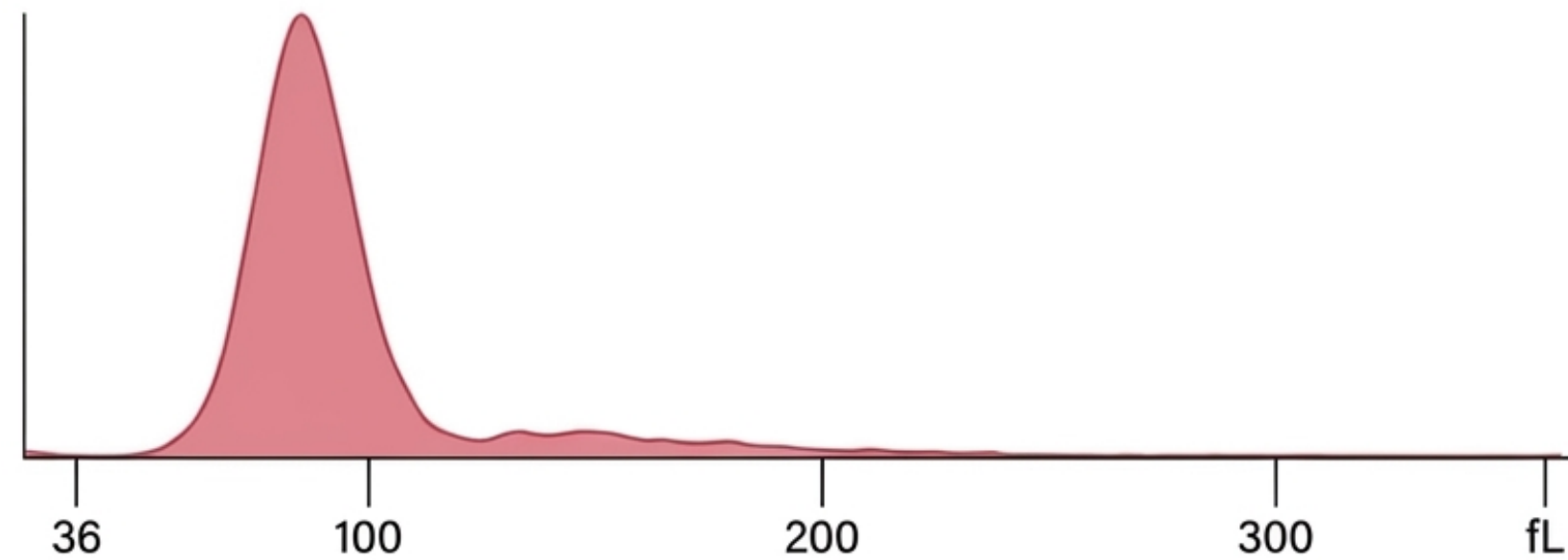


WBC: Impedance Count

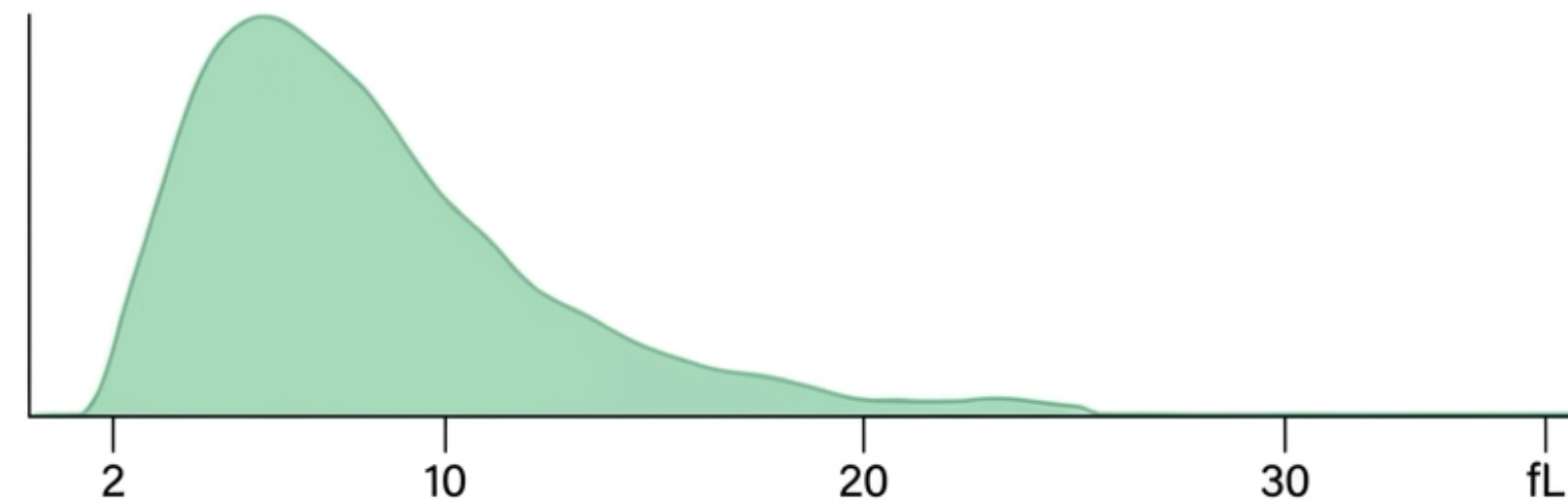


HGB: Spectrophotometry (525 nm)

RBC & PLT Chamber

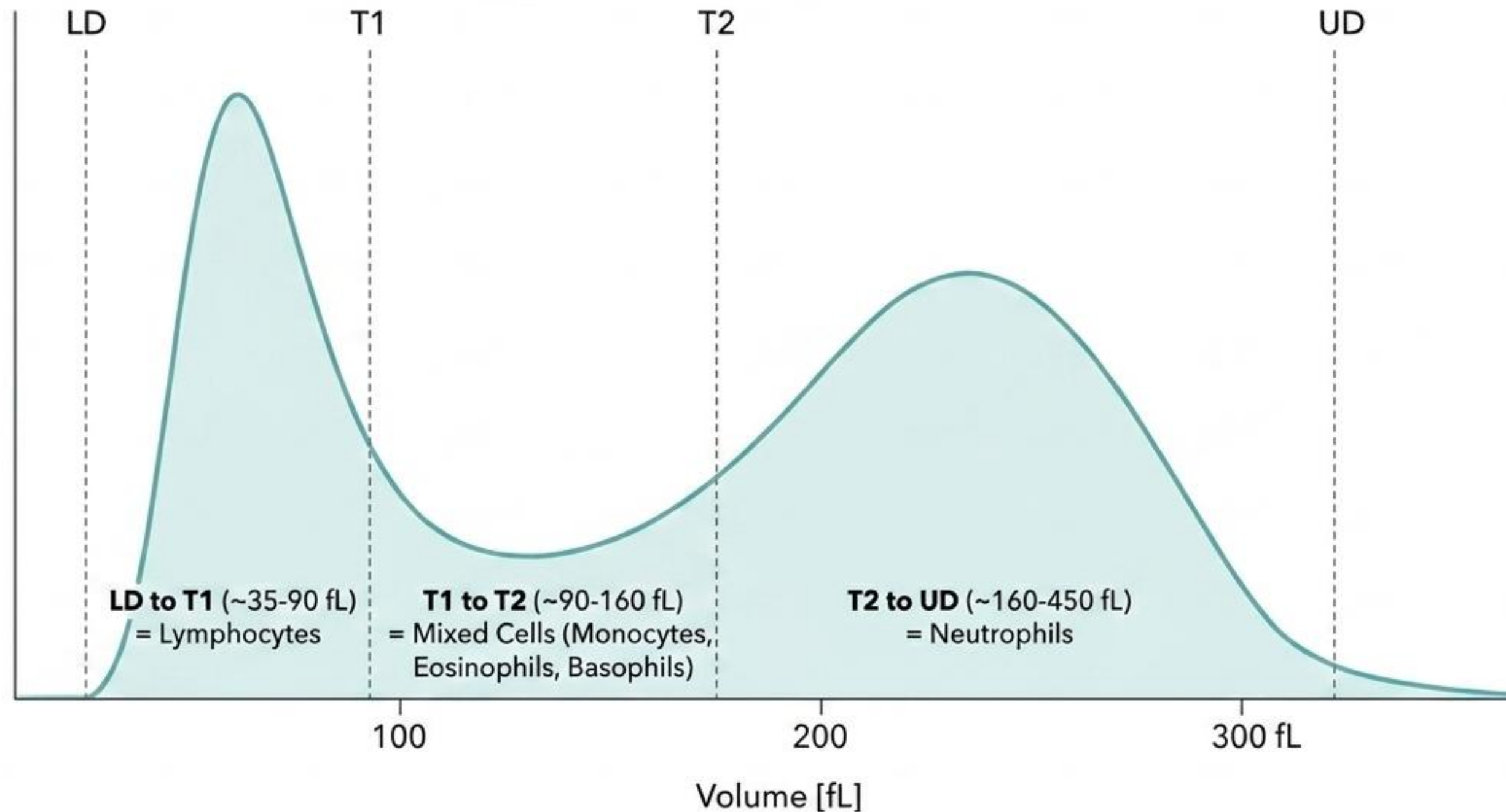


Particles 25–250 fL = Red Blood Cells (RBCs)



Particles 2–30 fL = Platelets (PLTs)

3-Part Differential: POC Systems



5-Part Differential: VCS Technology

V = Volume

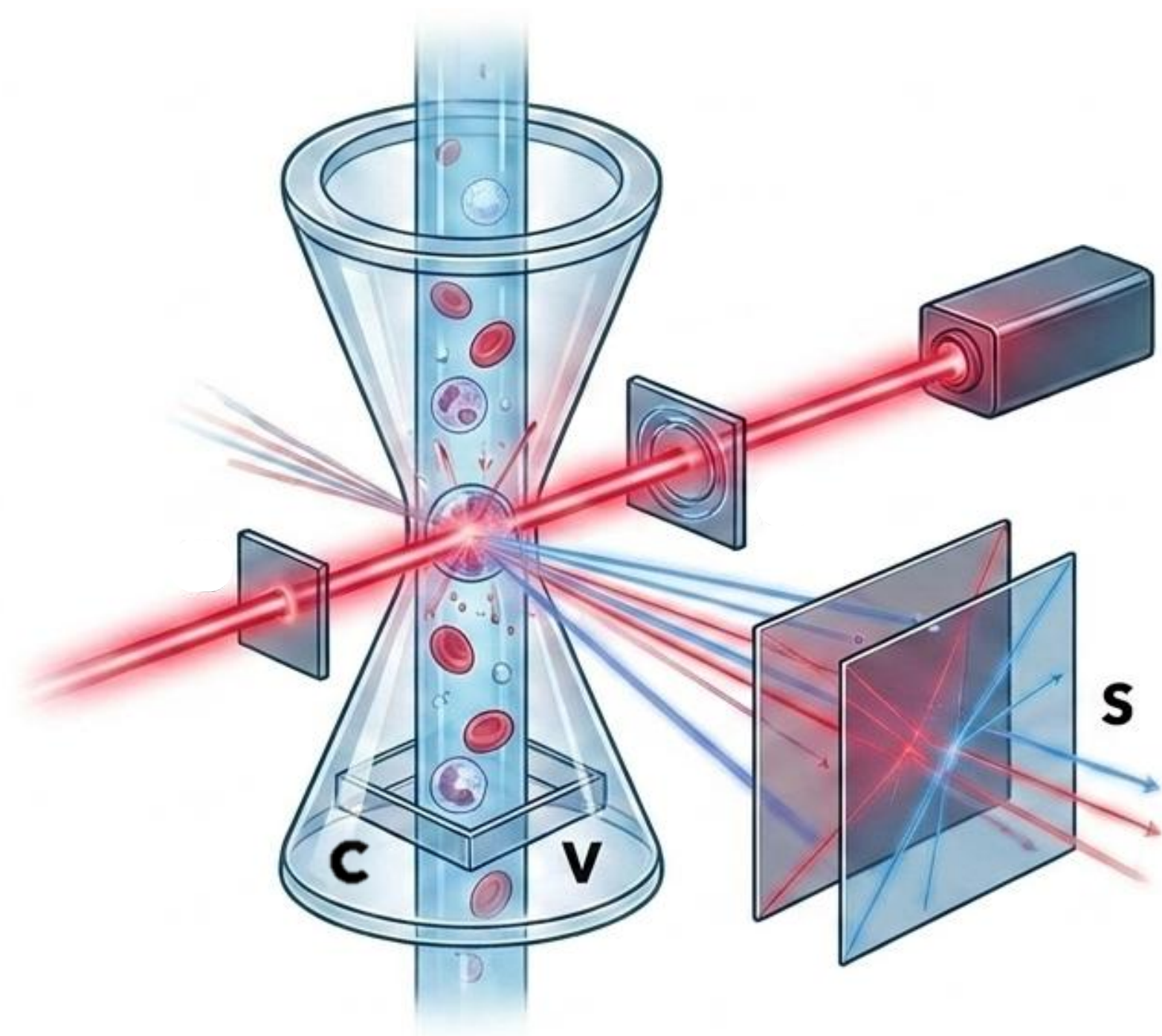
(Low-frequency Direct Current)

C = Conductivity

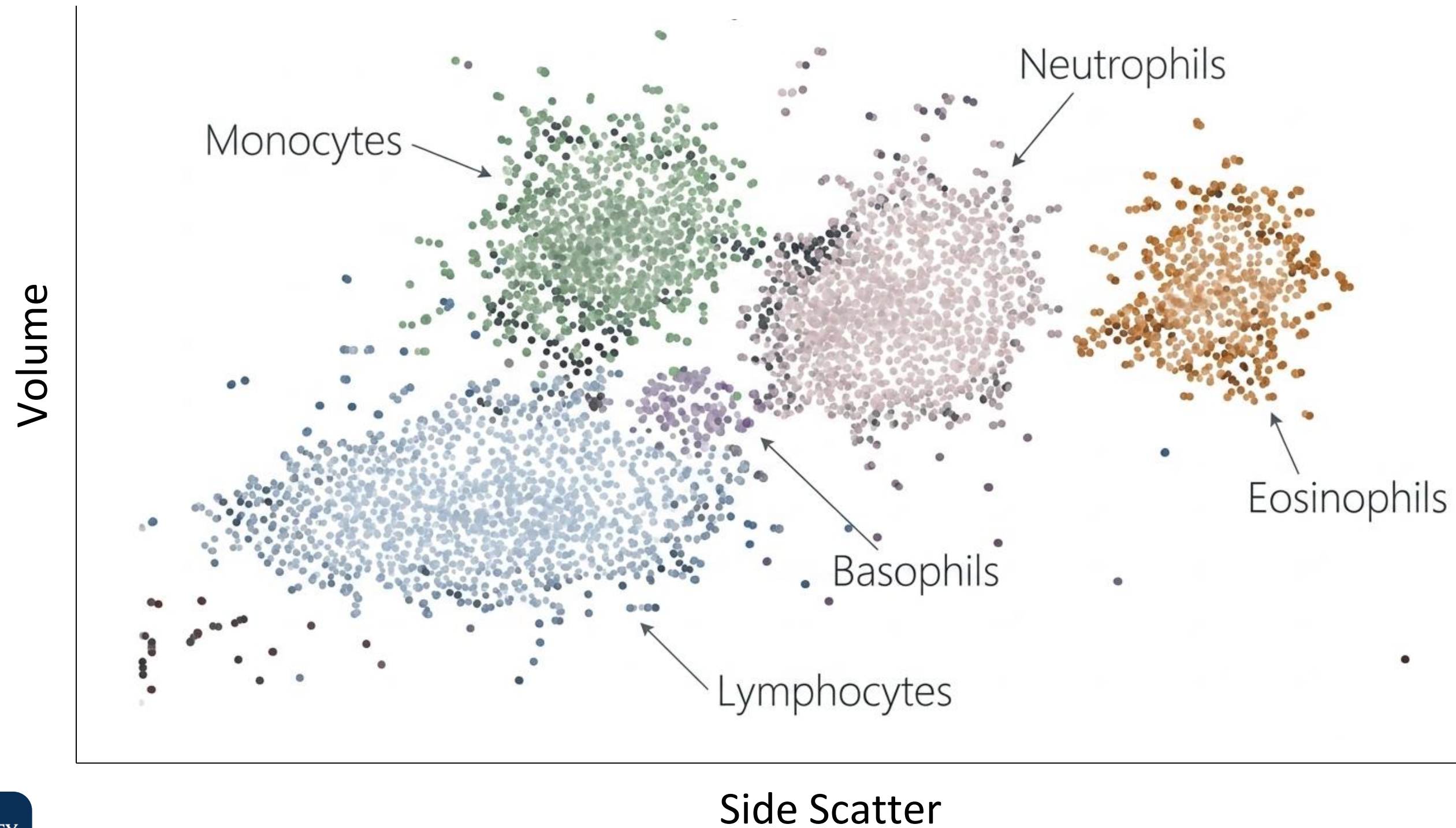
(High-frequency Radio Current
- maps internal contents)

S = Scatter

(Diode Laser - maps cellular
granularity/refractivity)



VCS Scatterplot



Error Codes & Parameter Flags

+ + + + +	Exceeds operating/linearity range
- - - - -	Total voteout (No histogram appears)
R	Review result (Recalculate unverified derived parameter)
? ? ? ? ?	Outside format display range
: : : : :	Flow cell clog detected
.	Incomplete computation

Maintenance & QC



Daily/Shift

Cleanings |
Background Checks
| Shift Control Runs



Periodic (Weekly/Monthly)

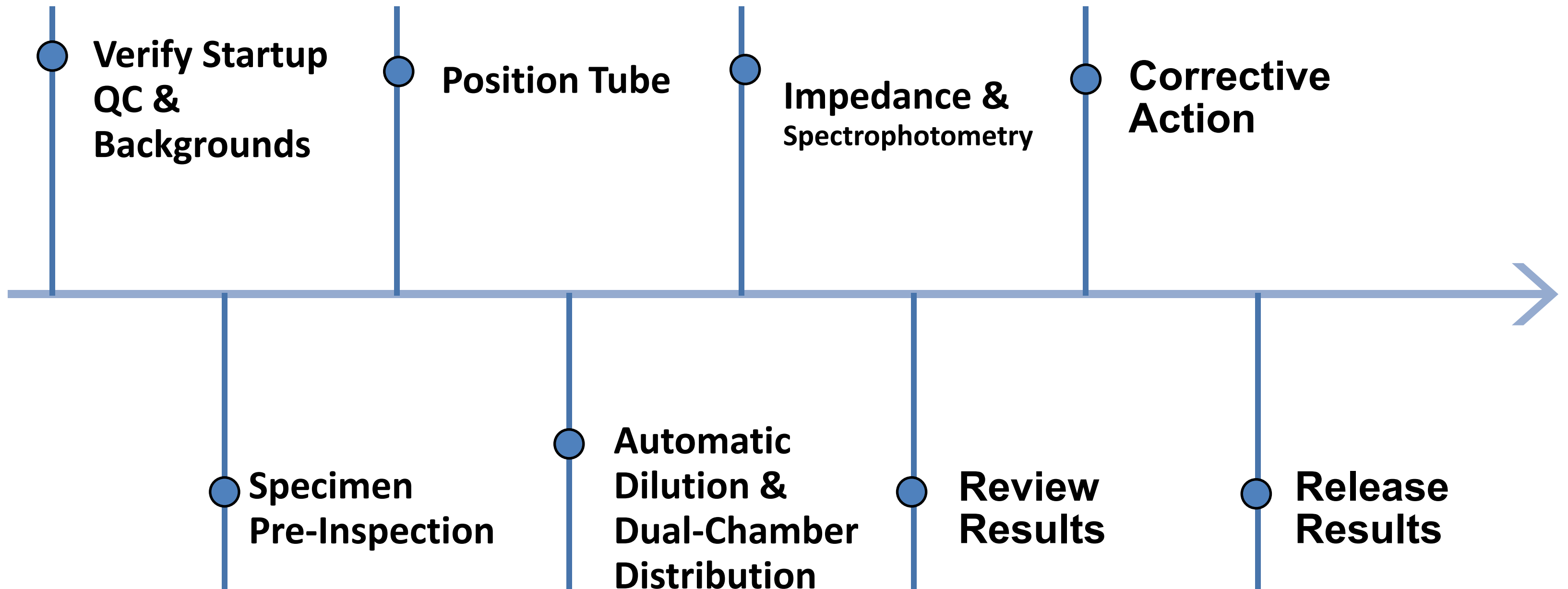
Deep Cleaning |
Preventative
Maintenance



Lot Validation

Daily Reagent
Verification |
Lot-to-Lot Reagent
Crossover

Analytical Procedure



Biosafety & Biohazard Controls



Standard Precautions